

For 1 kg of firewood burned, a standard inventory factor is **≈1.81 kg CO₂** emitted at the stack (plus very small CH₄ and N₂O).[\[engineeringtoolbox\]](#)

For 1 kg of fresh cow dung left to **decompose as a wet, ponded slurry** in a warm place like Assam, the dominant GHG is typically **methane (CH₄)**, and an IPCC-default style calculation gives a rough range of **~15–20 g CH₄ per kg fresh dung** (≈0.39–0.54 kg CO₂e on a 100-year basis).[ghginstitute+4](#)

Firewood: 1 kg burned

EPA's "Wood and Wood Residuals" factor is ~1,640 kg CO₂ per short ton, which converts to **≈1.81 kg CO₂ per kg wood**.[\[engineeringtoolbox\]](#)

EPA also lists ~126 g CH₄ and ~63 g N₂O per short ton of wood (≈0.14 g CH₄/kg and ≈0.07 g N₂O/kg).[\[engineeringtoolbox\]](#)

Cow dung: 1 kg left ponded (Assam)

A wet, ponded slurry is closer to IPCC manure categories like **Liquid/Slurry** (or in more anaerobic, long-residence cases, **Anaerobic lagoon**), which have much higher methane conversion factors (MCFs) than "dung left dry on pasture."[\[energy.sustainability-directory\]](#)

The IPCC Emission Factor Database (EFDB) lists default MCFs (warm, >25°C) of **65% for Liquid/Slurry** and **90% for Anaerobic lagoon**.[\[energy.sustainability-directory\]](#)

Worked methane estimate (with assumptions shown)

To express emissions "per kg fresh dung," you need volatile solids (VS) per kg fresh dung; one published dataset implies **VS ≈ 0.13 kg VS per kg fresh manure** (2.6 kg VS/day from ~20 kg manure/day).[\[journals.assaf.org\]](#)

Using an IPCC default maximum methane potential for cattle manure of **B₀=0.24 m³ CH₄ per kg VS**, the maximum CH₄ potential per kg fresh dung is

$0.13 \times 0.24 \approx 0.031$ m³ CH₄.prepp+1

Applying warm-climate MCFs then gives expected CH₄ formation of

$0.031 \times 0.65 \approx 0.020$ m³ (Liquid/Slurry) to $0.031 \times 0.90 \approx 0.028$ m³ (Anaerobic lagoon) per kg fresh dung.prepp+2

Converting volume to mass using methane density at STP (0°C, 1 atm) of **≈0.7168 kg/m³**, that is about **14.6 g CH₄/kg** (Liquid/Slurry) to **20.2 g CH₄/kg** (Anaerobic lagoon).[\[engineeringtoolbox\]](#)

Using IPCC AR6 100-year GWP for non-fossil (biogenic) methane of **~27.0**, this corresponds to roughly **0.39–0.54 kg CO₂e per kg fresh dung** (CH₄ only).[\[ghginstitute\]](#)

Important limits (so you don't over-interpret)

These are **Tier-1 style** estimates: real emissions will vary with how long the slurry stays ponded, temperature seasonality, crust formation, dilution with water, and whether gas bubbles escape or oxidize at the surface. [[energy.sustainability-directory](http://energy.sustainability-directory.org/)]

I didn't calculate N₂O from the ponded dung because converting to "per kg fresh dung" needs dung nitrogen content (or animal N-excretion), and anaerobic slurry conditions often shift the balance strongly toward CH₄ rather than N₂O anyway. drufire+1

If you tell me roughly how many days/weeks the slurry sits before drying out or being flushed away, I can suggest whether "Liquid/Slurry (65% MCF)" or "Lagoon (90% MCF)" is the closer match for your site.

For BioViL: (1. Project - Rampur, Assam)

At the farmers location:

- Monthly animal waste generated = 600 kg
- Monthly firewood burned = 120 kg

Hence, after the construction of the biodigester:

- GHGs generated by unutilized, decomposing and ideal animal waste: **218 kg CO₂e** per month = **2616 kg/a**

Calculated totals (600 kg, warm liquid/slurry, MCF=65%):

- CH₄ volume per kg fresh waste: $0.13 \times 0.24 \times 0.65 \approx 0.0203 \text{ m}^3 \text{ CH}_4/\text{kg}$.
- CH₄ mass per kg fresh waste: $0.0203 \times 0.7168 \approx 0.0145 \text{ kg CH}_4/\text{kg}$ ($\approx 14.5 \text{ g/kg}$).
- Total CH₄: $600 \times 0.0145 \approx 8.7 \text{ kg CH}_4$.

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- CO₂e (using EPA's CH₄ GWP₁₀₀=25 for consistency with the wood calculation): $8.7 \times 25 \approx 218 \text{ kg CO}_2\text{e}$.

- GHGs generated due to burning of wood: **220.1 kg CO₂e** per month = **2641.2 kg/a**

Calculated totals (120 kg wood):

- CO₂: $120 \times 1.81 \approx 217.2 \text{ kg CO}_2$.
- CH₄: $120 \times 0.14 \text{ g/kg} \approx 16.8 \text{ g CH}_4$ (0.0168 kg).
- N₂O: $120 \times 0.07 \text{ g/kg} \approx 8.4 \text{ g N}_2\text{O}$ (0.0084 kg).

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- CO₂e using EPA GWPs (CH₄=25, N₂O=298): $217.2 + 0.0168 \times 25 + 0.0084 \times 298 \approx 220.1 \text{ kg CO}_2\text{e}$.